
The emergence of visual simulation in task-optimized recurrent neural networks

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Abstract

1 Primates display remarkable prowess in making rapid visual inferences even when
2 sensory inputs are impoverished. One hypothesis about how they accomplish
3 this is through a process called visual simulation, in which they imagine future
4 states of their environment using a constructed mental model. Though a growing
5 body of behavioral findings, in both humans and non-human primates, provides
6 credence to this hypothesis, the computational mechanisms underlying this ability
7 remain poorly understood. In this study, we probe the capability of feedforward
8 and recurrent neural network models to solve the *Planko* task, parameterized to
9 systematically control task variability. We demonstrate that visual simulation
10 emerges as the optimal computational strategy in deep neural networks only when
11 task variability is high. Moreover, we provide some of the first evidence that
12 information about imaginary future states can be decoded from the model latent
13 representations, despite no explicit supervision. Taken together, our work suggests
14 that the optimality of visual simulation is task-specific and provides a framework
15 to test its mechanistic basis.

16 1 Introduction

17 A longstanding goal in the brain sciences is to understand the neural algorithms and computations
18 that support humans’ ability to interact optimally with their surroundings. A popular cognitive level
19 theory for how humans visually reason about their environments, under uncertainty, is that they
20 rely on “simulation” through rich internal generative models of the world Kersten & Yuille (1996);
21 Tenenbaum et al. (2011); Battaglia et al. (2013); Ullman et al. (2017) to build and test hypotheses
22 about the future and plan effective behavior. The notion of visual simulation has been discussed
23 since at least Descartes, who theorized that this ability is implemented in the brain through the
24 same neural mechanisms as perception, and operates without any stimulation from the external
25 world Lokhorst (2005). Ullman expanded upon this theory in his seminal Visual Routines (Ullman
26 (1984)), in which he suggested that in order for visual simulation to work effectively it must utilize
27 syntactic computations, which can be flexibly re-applied to any visual features. Recent studies in
28 humans and non-human primates have provided insight into the potential neural underpinnings of
29 these cognitive-level theories Ahuja et al. (2022); Ahuja & Sheinberg (2019); Rajalingham et al.
30 (2021, 2022). While Ahuja & Sheinberg (2019) demonstrated the ability of simple feedforward
31 neural networks (FFNs) to perform their visual simulation task, they find a misalignment between
32 model and primate behavior. Similarly, Rajalingham et al. (2021) show this misalignment in recurrent
33 neural networks (RNNs) trained to play a simplified version of Pong (*M-Pong*) where the RNN had
34 to guess where to move a paddle to catch a linearly-moving ball. However, the authors found that
35 the same RNNs, when trained to predict the position of M-Pong balls across their trajectories, were

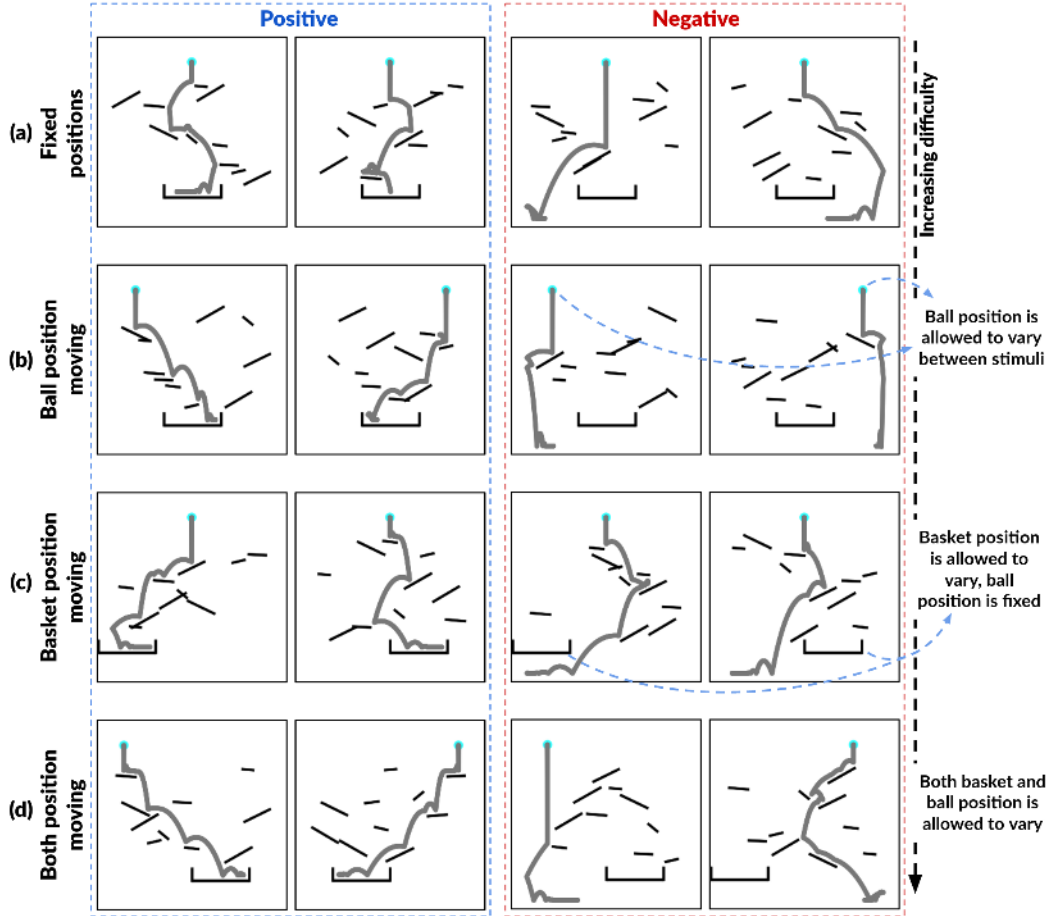


Figure 1: **Planko as a visual simulation task.** Observers are shown a single image of a game board and asked to predict whether the ball at the top of the image (in light blue) will fall into the catcher at the bottom of the image. The grey lines in each image depict the trajectory of the ball at the top, simulated in a world with Newtonian physics. We generate positive and negative game boards for every ball and cup position, in which the planks are placed in a way to bounce the ball into the catcher or not. We also generate four different versions of the game, in which the properties of the ball and cup are modified: (a) Fixed positions of each, (b) the ball position is randomly sampled, (c) the bucket position is randomly sampled, and (d) both the ball and bucket positions are randomly sampled. Each modification increases the total number of game boards that can be generated, and hence, the game’s difficulty.

36 able to learn routines for visual simulation that explained significantly more variance in behavior and
 37 neural activity than RNNs without this constraint.

38 **Contributions** We explore the conditions under which visual simulation naturally emerges as the
 39 optimal computational strategy in deep neural networks purely driven by task-constraints. We refrain
 40 from providing any source of information about temporal dynamics to our network models, for
 41 example, watching object trajectories or explicit supervision about the locations of objects in the
 42 world. We start with the visual simulation task developed by Ahuja & Sheinberg (2019) and adapt it
 43 to our suite of models and call it *Planko* (Figure. 1). In *Planko*, observers are tasked with predicting
 44 the outcome of a ball falling through a random series of oriented planks without ever seeing the ball’s
 45 trajectory. Unlike *M-Pong*, *Planko* is parameterized to make it possible to generate game boards that
 46 range from trivially easy to extremely difficult. We investigate whether models can learn to solve
 47 *Planko*, and whether the solutions they learn resemble the Newtonian physics used to generate game
 48 boards despite having no explicit access to that information.

- We find that a variety of feedforward deep neural networks (FFNs) and RNNs learn accurate solutions for easy versions of *Planko*, but only an attentional circuit model grounded in neurobiology can solve harder versions of the task (InT, Linsley et al. 2021).
- The InT’s attention maps indicate that it learns to focus on paddles that may interact with the *Planko* ball, and regions of the game board where it expects the ball to fall through.
- A decoding analysis demonstrated that the InT incrementally simulates a *Planko* ball’s trajectory through the game board, in hard but not easy game boards, and that this path closely approximates the ground-truth trajectory generated in each board with Newtonian physics.
- Our findings indicate that robust visual simulation emerges as an optimal algorithm in difficult environments, and that prior work Rajalingham et al. (2021, 2022) suggesting that additional learning constraints are needed for visual simulation may be a byproduct of a trivial task driving models to learn shortcuts.

2 The *Planko* challenge

The *Planko* challenge is inspired by prior work in visual simulation, which measured primate accuracy in simulating the trajectory of moving balls, and used fMRI to identify regions of cortex that correlated with their behavior (Ahuja et al., 2022). Much like *M-Pong*, models trained on that task learned shortcut solutions to solve it (Ahuja & Sheinberg, 2019). With our *Planko* challenge, we have controlled for variations in the task space to explicitly prevent the learning of shortcut solutions, and in order to understand the extent to which it changes the strategies learned by models for visual simulation.

Each *Planko* board depicts a ball at the top of the screen placed above ten randomly oriented and positioned planks. A catcher is placed at the bottom of the screen (Figure. 1). Each plank is parameterized by its angle of inclination, length, and its position on the screen. The *Planko* ball and catcher are placed in accordance with task difficulty as discussed below. The physics of this world are specified by Newton Dynamics (<http://www.newtondynamics.com>). The ball’s trajectory as

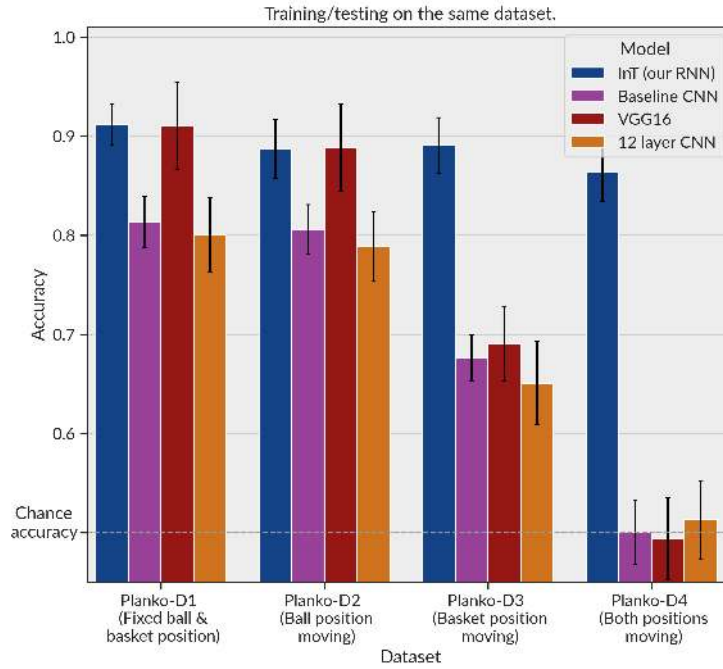


Figure 2: **FFN and RNN performance on the *Planko* challenge.** Error bars depict 95% bootstrapped confidence intervals. The InT is significantly more accurate than any other model on the most challenging versions of *Planko*: when the basket position or both the basket and ball positions are randomly placed across stimuli.

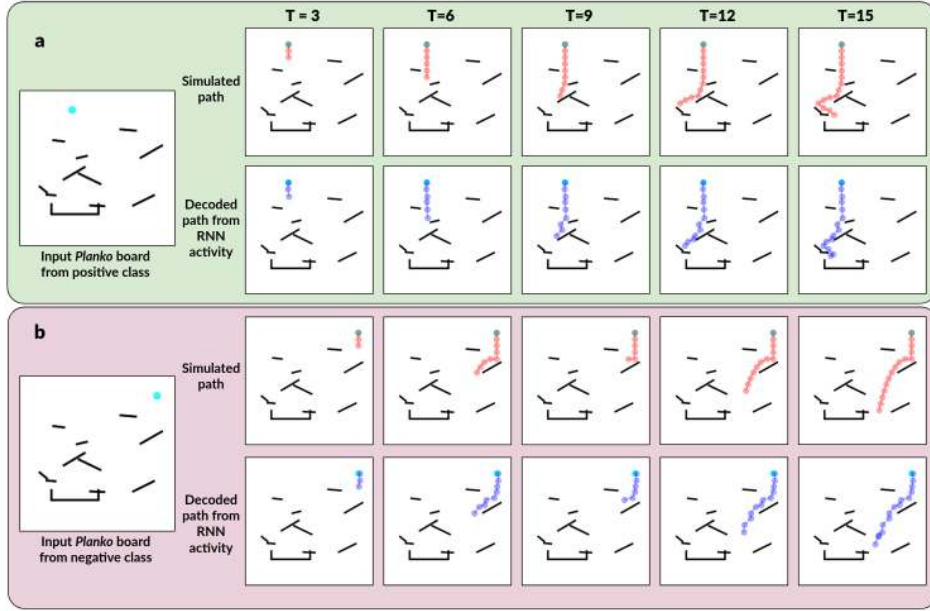


Figure 3: **The InT learns to solve *Planko* by learning a visual simulation strategy that resembles the ground truth physics.** Sample positive and negative *Planko* boards were shown to an InT while decoding the position of the ball from the activities of excitatory units. Ground truth paths are depicted in red and decoded paths from the InT are depicted in blue.

it falls downward is tracked to determine if it eventually lands in the catcher (positive class) or falls to the ground (negative class). Figure 1 illustrates example *Planko* ball trajectories for both positive and negative classes.

2.1 Parameterizing *Planko* board difficulty

By bounding the variations in the *Planko* board elements, we systematically control for the challenge associated with solving the board for neural network models. *Planko*-D1 (Figure. 1a) is the easiest variant of the task in which both the ball and catcher positions are constant across the entire dataset. *Planko*-D2 (Figure. 1b) and *Planko*-D3 (Figure. 1c) are intermediate-level boards. While in *Planko*-D2 the initial ball position is randomly sampled from the upper 40% of the game board (with the catcher position remains constant), *Planko*-D3 places the catcher in a random location sampled from the lower 40% of the game board (with the ball position constant). *Planko*-D4 (Figure. 1d) is the version of the task wherein both the ball and catcher positions are stochastic. Boards in which the ball hits either vertical wall are excluded from the data used for the neural network analysis.

3 RNNs, but not FFNs, solve *Planko*-D4

General setup All models used herein were trained to classify each *Planko* game board into one of either positive or negative classes. Model parameters were optimized with Stochastic Gradient Descent implemented via the Adam algorithm Kingma & Ba (2014) with an initial learning rate of $3e - 4$. Binary Cross Entropy (BCE) was used as the training objective. Each train (test) dataset consisted of 200K (5K) *Planko* boards of dimensions 64×64 pixels. Training was carried out on a NVIDIA TITAN Xp GPU for 100 epochs while measuring validation accuracy after each epoch over a held-out set of 10K boards.

The InT Model The Index-and-Track circuit (a complete model description in Linsley et al. 2021) architecture consisted of an input layer with $64 \ 1 \times 1$ convolutional filters followed by the InT circuit with 3×3 horizontal kernels and 64 output channels. A 1×1 convolutional “readout” followed by a linear layer transformed the final RNN hidden state to the classification output. The RNN is trained for $T = 24$ time steps.

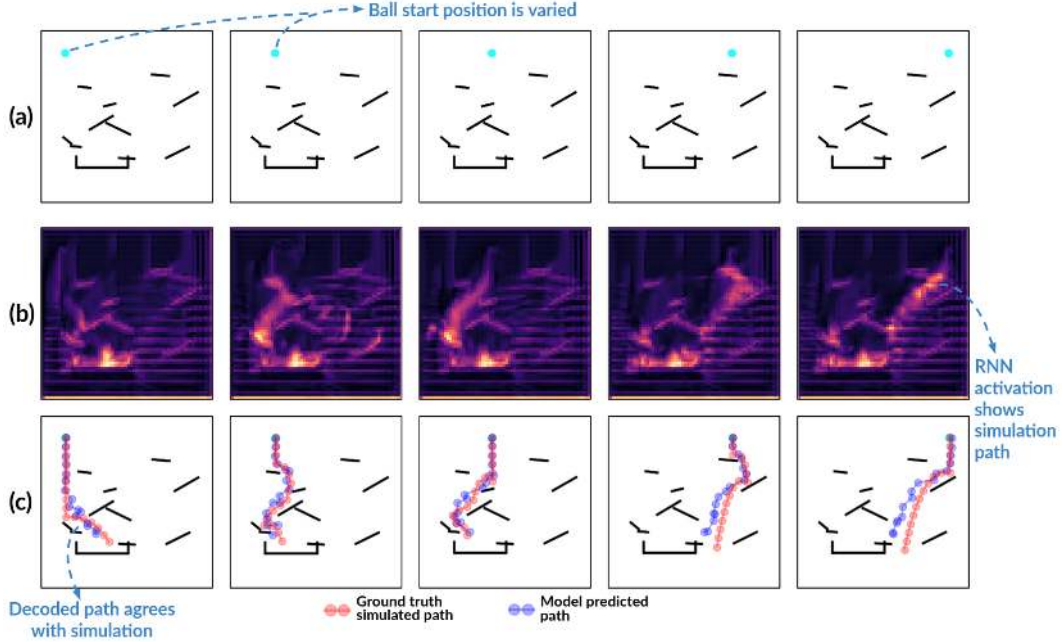


Figure 4: **The InT’s attention maps reveal solution strategies.** (a) Testing the RNN (InT) with boards where everything is held constant except the ball which is horizontally translated. (b) The hidden state activity at timestep T for the RNN for the boards in (a). (c) The decoded ball positions from the hidden state vs the ground truth simulated paths.

100 **Baselines** In addition to the InT, we trained a simple feedforward 2-layer convolutional neural network
 101 (termed “Baseline CNN”), a standard VGG16 Simonyan & Zisserman (2014), and a 12-layer CNN
 102 with a parameter count identical to the InT circuit.

103 **Classification results** The performance landscape of models across the *Planko* tasks revealed that
 104 both FFNs and RNNs solved easier versions of the task (Figure. 2). However, the InT was significantly
 105 more accurate on *Planko*-D4, the most challenging version.

106 **Decoding analysis** We train a decoder to extract the coordinates of the *Planko* ball positions from
 107 the final timestep InT activities. The decoder is trained to minimize the mean-squared error between
 108 the predicted ball coordinates and the ground truth ball position obtained from the physics simulator.
 109 The decoder consisted of three layers of 1×1 convolution and pooling operations and finally a linear
 110 readout layer. The model was trained on 64 channel 64×64 feature tensors from the final timestep
 111 of the trained InT circuit. A total of 16 decoder models are trained for each of the 16 ball positions
 112 from the simulator for 20 epochs with 200,000 feature tensors. The mean-squared error is measured
 113 on the validation set after every epoch and the model with the least error is used to predict the ball
 114 position from new boards.

115 4 Conclusion

116 We explore the conditions under which “visual simulation” emerges as the naturally optimal algorithm
 117 in task-optimized RNNs. We demonstrate that only the most performant RNN, on our most variable
 118 task, adopts a “simulation” strategy. To the best of our knowledge, we provide the first evidence that
 119 information about imaginary future states can be decoded from RNN internal representations. While
 120 this work is preliminary, we are hopeful that it paves the way for RNN-guided electrophysiology
 121 research to understand the mechanistic basis of visual simulation.

Checklist

The checklist follows the references. Please read the checklist guidelines carefully for information on how to answer these questions. For each question, change the default **[TODO]** to **[Yes]**, **[No]**, or **[N/A]**. You are strongly encouraged to include a **justification to your answer**, either by referencing the appropriate section of your paper or providing a brief inline description. For example:

- Did you include the license to the code and datasets? **[Yes]** See Section ??.
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1. For all authors...

- (a) Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope? **[Yes]**
- (b) Did you describe the limitations of your work? **[Yes]**
- (c) Did you discuss any potential negative societal impacts of your work? **[N/A]**
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2. If you are including theoretical results...

- (a) Did you state the full set of assumptions of all theoretical results? **[N/A]**
- (b) Did you include complete proofs of all theoretical results? **[N/A]**

3. If you ran experiments...

- (a) Did you include the code, data, and instructions needed to reproduce the main experimental results (either in the supplemental material or as a URL)? **[No]**
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- (c) Did you report error bars (e.g., with respect to the random seed after running experiments multiple times)? **[Yes]**
- (d) Did you include the total amount of compute and the type of resources used (e.g., type of GPUs, internal cluster, or cloud provider)? **[Yes]**

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- (a) If your work uses existing assets, did you cite the creators? **[Yes]**
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- (e) Did you discuss whether the data you are using/curating contains personally identifiable information or offensive content? **[N/A]**

5. If you used crowdsourcing or conducted research with human subjects...

- (a) Did you include the full text of instructions given to participants and screenshots, if applicable? **[N/A]**
- (b) Did you describe any potential participant risks, with links to Institutional Review Board (IRB) approvals, if applicable? **[N/A]**
- (c) Did you include the estimated hourly wage paid to participants and the total amount spent on participant compensation? **[N/A]**

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